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Performance, techno-economic viability, and environmental impact of Domestic Solar Tri-Generation System (DSTS): A comparative study of copper and galvanized iron-based systems for sustainable building applications

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A R T I C L E I N F O

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A B S T R A C T

Research on building-integrated solar concrete water heaters has largely focused on single-function applications, with limited emphasis on multifunctional systems combining water heating, electricity generation, and cooling. Conventional photovoltaic–thermal (PVT) and building-integrated PVT (BIPVT) systems face challenges such as complex integration, seasonal efficiency fluctuations, added costs, and limited passive cooling. To address these limitations, this study presents a Domestic Solar Tri-Generation System (DSTS) that integrates solar water heating, photovoltaic (PV) power generation, and thermoelectric cooling within a concrete slab serving as a building roof. Two variants-a copper-based DSTS (C-DSTS) and a galvanized iron-based DSTS (GI-DSTS), were constructed and tested under the climatic conditions of Vellore, India, at flow rates of 0.12, 0.24, and 0.36 L/min. The systems achieved overall efficiency of 46.4 %, with thermal efficiencies of 24.75 % (C-DSTS) and 24.20 % (GI-DSTS), and exergy efficiencies of 37.05 % and 32.30 %, respectively. Passive indoor cooling of 6.2 °C (C-DSTS) and 5.9 °C (GI-DSTS) was recorded, while 108 L of water was heated to 50 °C within 5 h. Annual electricity generation reached 198 kWh, with lifetime savings of 55,113.57 USD and CO_2 mitigation of 180,704–206,921 kg, alongside carbon credit earnings of over 4000 USD. Unlike conventional BIPVT-TEG systems, the DSTS embeds thermoelectric modules within the slab, stabilizing temperature gradients, enhancing passive cooling, and eliminating auxiliary devices. This multifunctional, modular, and durable design shows strong potential for integration in residential buildings, providing a scalable and cost-effective pathway toward net-zero energy construction.

Abbreviation:

ALCC	Annualized Life Cycle Cost	AC	Air Conditioner
LCC	Life Cycle Cost	C-PVC pipe	Chlorinated-Poly Vinyl Chloride pipe
DSTS	Domestic Solar Tri-generation System	DAQ	Data Acquisition System
PV	Photovoltaic	HUF	Heat Utilization Factor
TEG	Thermo Electric Generator	COP	Coefficient of Performance

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(*continued*)

C-DSTS	Copper-Domestic Solar Tri-generation System	C-heater	Copper-heater
GI-DSTS	Galvanized Iron- Domestic Solar Tri-generation System	GI-heater	Galvanized Iron-heater
Nomenclature:			
$\dot{A}$	Collector surface area	m	Mass flow rate
$\mathscr{A}_c$	Cumulative savings	$\mathscr{n}$	Life time of DSTS
$\mathscr{A}_d$	Degradation adjusted annual savings	ns	Sunny days in a year
$\mathscr{A}_i$	Inflation adjusted annual savings	Nu	Nusselt number
$\mathscr{A}_{initial}$	Initial annual saving	$\mathscr{P}$	Payback period
$\mathscr{C}_{cooling}$	Annual energy cost avoided by replacing electric cooling systems	Pr	Prandtl number
$\mathscr{C}_{EWH}$	Annual energy cost avoided by replacing electric water heater	$\mathscr{PV}$	Present value of savings
$\mathscr{C}_i$	Initial investment cost	$Ppump$	Pump's Power consumption
$\mathscr{C}_{m,t}$	Maintenance cost	Qc	Thermal energy utilized for space cooling
$\mathscr{C}_{o,t}$	Operation cost	Qo	Useful heat gained by water
Cp	Specific heat capacity	Ql	Heat loss from the collector
$\mathscr{C}_{PV-TEG}$	Annual savings from the electricity generated by PV and TEG	Qh	Thermal energy utilized for water heating
$\mathscr{C}_{rf}$	Capital recovery factor	r	Interest rate
$\mathscr{C}_{r,t}$	Replacement cost	R_e	Reynolds number
$\mathscr{C}_s$	Salvage value	sd'	Standard deviation
$\mathscr{d}$	Degradation rate	$tout$	Outlet water temperature
Df_n	Discount factor	tin	Inlet water temperature
Df_t	Discount factor	ta	Ambient temperature
DT	Daily thermal energy output of DSTS	ts	Collector surface temperature
$\hat{E}$	Embodied energy	$\widehat{T}$	Solar exposure time
Ei	Exergy in	$u(i1)$ $u(i2)$ and $u(in)$	Uncertainties of independent parameters
Eo	Exergy out	$\overline{\overline{U}}$	Overall Heat Transfer Coefficient
$\hat{E}T$	DSTS's annual thermal energy output	v	Fluid velocity
$\acute{F}$	Efficiency factor	$\dot{V}$	Panel voltage
$\overline{G}$	Solar Irradiance	$\acute{\eta}T$	Thermal efficiency
hc	The convective heat transfer coefficient	$\acute{\eta}E$	Exergy efficiency
i	Inflation rate	$\acute{\eta}P$	Photovoltaic efficiency
$\dot{I}$	Panel current	ρ	Fluid density
$kconcrete$	Concrete's Thermal conductivity	μ	Dynamic viscosity
kl	Fluid's Thermal conductivity	∂C	Calculated parameter affected by independent parameters
kp	Pipe Thermal conductivity		
$Lconcrete$	Concrete length		
Lp	Pipe length		

1. Introduction

The rising energy demand for water heating and residential cooling underscores the need for sustainable solutions for domestic buildings. Globally, residential structures consume approximately 21 % of total energy, with water heating and cooling accounting for a significant share [1]. Specifically, water heating contributes to approximately 18 % of residential energy use, while cooling systems account for nearly 20 % [2]. In India, rapid urbanization and economic growth have increased domestic energy needs, with 6–8 % of household energy dedicated to water heating and approximately 10 % to cooling via fans and air conditioners (ACs) [3]. Notably, fans account for nearly 40 % of residential electricity use, while ACs, although used by fewer households, significantly impact peak power demand. Projections indicate that by 2030, the energy consumption for air conditioning and electric water heating could increase fivefold, with the overall residential energy demand expected to double by 2040 [4]. This surge has challenged grid stability and accelerated carbon emissions, necessitating innovative energy solutions. Photovoltaic thermal (PVT) systems have emerged as promising alternatives. One such solution is a solar concrete water heater, a Building-Integrated Photovoltaic Thermal (BIPVT) system that embeds solar-absorptive materials within concrete slabs to harness solar energy for water heating. This dual-purpose design enhances thermal efficiency while serving as a structural component of a building [5]. By reducing heat transmission indoors, it improves cooling and simultaneously addresses water heating needs. Integration with cooling technologies further minimizes reliance on conventional energy, lowers electricity costs and emissions, and makes solar concrete water heaters highly suitable for sun-rich regions.

To evaluate their effectiveness, it is essential to compare solar concrete water heaters with conventional PVT systems. These systems, including solar concrete water heaters, are categorized based on their cooling methods and structural integration. Air-based PVTs extract heat via airflow for ventilation and space heating [6], whereas water-based PVTs use liquid as a heat transfer medium, offering higher thermal efficiency through flat plates, glazed, or unglazed collectors [7]. Hybrid PVTs combine air and water cooling for improved heat dissipation. BIPVT systems serve as part of the building envelope, reducing the material costs while generating electricity and thermal energy [8]. Concentrated PVTs (CPVTs) enhance the power output using optical elements, but require advanced cooling [9]. Phase Change Material (PCM)-integrated PVTs store thermal energy to regulate temperature and boost efficiency [10]. Thermoelectric PVTs (TEG-PVTs) incorporate TEGs to convert waste heat into electricity via the Seebeck effect, thereby improving the energy generation [11]. A comparative analysis of conventional PVT systems is shown in Table 1.

Table 1
Technical comparative analysis of DSTS and conventional PVT systems.

Feature	Air-Based PVT [6,12–14]	Water-Based PVT [7,10,15]	Hybrid PVT [8,16,17]	BIPVT [9,18–20]	TEG-PVT [11,21,22]	Proposed DSTS
Structural Integration	Mounted on roof or walls, requiring additional support structures	Mounted on roof with water piping, requiring separate heat exchanger	Requires ductwork and piping, increasing weight and complexity	Integrated into roof/façade, reducing material redundancy	PV panels with TEGs installed separately, increasing load and material costs	Fully embedded within the concrete slab, serving as a load-bearing structure while integrating PV, TEGs, and thermal absorbers
Space Utilization	Requires dedicated rooftop or façade space, reducing available area for other installations	Requires storage tanks, circulation pipes, and mounting reinforcements	Requires dual-channel flow paths, complicating design and installation	Optimizes space by replacing traditional roofing/wall materials, maintaining building aesthetics	Increases material footprint due to layered TEG-PV assembly	Maximizes space efficiency by eliminating separate mounting, allowing flexible placement over selected roof areas
Cooling Mechanism	Uses forced or natural convection to dissipate heat; effectiveness depends on airflow velocity and ambient conditions	Uses forced circulation of water with heat exchangers; efficiency depends on water flow rate and heat exchanger design	Combines air and water for enhanced thermal extraction, improving PV efficiency	Building materials act as thermal insulators, stabilizing temperature variations	TEGs extract heat via the Seebeck effect, but require additional cooling to maintain efficiency	Concrete slab + TEGs + embedded water pipes enable multi-stage heat dissipation, improving thermal efficiency by passive cooling via embedded TEGs
Passive Cooling Benefit	Limited, airflow depends on wind speed and ambient temperature fluctuations	Moderate, heat absorption prevents overheating but requires active pumping	Moderate to high, as dual cooling mechanisms increase heat transfer rate	Moderate, insulation slows heat penetration but does not actively remove heat	Minimal, TEGs reduce PV temperature but provide no large-scale cooling effect	Up to 6.2 °C indoor temperature reduction due to heat absorption, redistribution, and passive cooling via TEGs and water channels
Energy Output Efficiency	PV efficiency drops by ~0.4–0.5 % per °C increase in temperature	Higher thermal efficiency (40–60 %), but electrical efficiency drops due to heat stress	Optimized dual energy conversion, but requires careful fluid flow control	Improved thermal performance due to building envelope integration	TEGs convert ~5–8 % of waste heat into electricity, but power output depends on ΔT stability	Maximizes PV efficiency by preventing thermal degradation, utilizes waste heat for both electricity generation and space cooling
Installation Complexity	Requires ducting, ventilation channels, and proper air circulation management	Requires fluid circulation system, heat exchanger, and storage tank, increasing costs and weight	Requires combined air and water circulation network, making system integration difficult	Requires specialized roofing/wall integration techniques but reduces secondary material costs	Requires TEG attachment layers, heat sinks, and thermal interface materials, increasing design complexity	Pre-integrated system, reducing installation time, component redundancy, and maintenance requirements
Maintenance Requirement	Filters and ducts require periodic cleaning; airflow sensors and fans degrade over time	Piping system prone to leaks, corrosion, and scaling, requiring periodic descaling and pump maintenance	Higher maintenance due to dual circulation system	Low maintenance, as system is embedded within building materials	TEGs degrade over time, with efficiency loss due to material fatigue	Minimal maintenance, as components are encased within concrete, preventing exposure to environmental damage
Thermal Energy Storage	No inherent heat storage, relies on continuous airflow	Stores heat in water, improving system reliability but requiring large insulated tanks	Stores heat in both air and water, offering flexible utilization	Some thermal retention in building elements, depending on material selection	No significant storage, as TEGs primarily convert heat to electricity	Concrete slab acts as a thermal buffer, storing and gradually releasing heat for better temperature regulation
Indoor Comfort Improvement	No impact, as system operates independently of indoor conditions	Minimal, only indirect benefits from reduced rooftop heat transfer	Moderate, dual cooling reduces indirect heat gain	Moderate, enhances insulation but does not actively cool	Limited, as TEGs primarily focus on power generation, not space cooling	Significantly improves indoor comfort, reducing roof heat penetration and cooling loads
Long-Term Durability	Moderate, dependent on air duct material quality and fan reliability	Lower lifespan due to corrosion and scaling of pipes and tanks	Moderate, requires maintenance to prevent dual-system failures	High durability, as system is part of the building envelope	Lower lifespan due to TEG degradation and thermal fatigue	Extremely high durability, as PV and TEGs are protected within concrete, minimizing environmental wear and tear
Cost-Effectiveness	Moderate, additional ducting and ventilation control increase costs	High initial investment, with costs for tanks, pumps, and insulation [Ref-20]	Expensive, due to dual-system complexity and material use	Moderate, reduces material costs by replacing conventional roofing/wall elements	Expensive, as TEGs require multiple interface layers, heat sinks, and advanced materials	Lower material and maintenance costs, with long-term energy savings, reduced carbon footprint, and lower lifecycle costs

Table 2
Comprehensive literature review table on BIPVT systems.

BIPVT Type	Building Integration	Studied Parameter	Assessment Parameters	Thermal Efficiency (%)	Electrical Efficiency (%)	Exergy Efficiency (%)	Cost Analysis ($/m^2 or Payback Period in Years)	Environmental Impact (CO_2 Reduction, Carbon Payback Time, etc.)	Major Findings	Remarks	Reference
Air-Based BIPVT	Roof, Wall, Façade	Electrical & thermal efficiency	PV temperature reduction, ventilation rate, heat extraction efficiency	30–50 %	10–16 %	8–14 %	$200–300/m^2, Payback: 6–8 years	15–30 % CO_2 reduction	Air-based BIPVTs can reduce PV temperature but have lower heat capacity than water-based systems	Seasonal limitations: not effective in extreme summers or winters	[20,23]
Water-Based BIPVT	Roof, Wall	Thermal performance, electrical efficiency	Water temperature, heat transfer coefficient, PV efficiency	50–70 %	12–18 %	10–20 %	$250–400/m^2, Payback: 5–7 years	20–40 % CO_2 reduction	Higher thermal efficiency than air-based systems; risk of freezing in cold climates	Needs antifreeze measures for cold regions	[24]
Hybrid (Air + Water) BIPVT	Roof, Wall	Combined cooling and heating performance	Dual heat extraction efficiency, energy savings	55–75 %	14–19 %	12–22 %	$300–450/m^2, Payback: 5–6 years	25–45 % CO_2 reduction	Optimized for seasonal variations; offers higher overall efficiency	Complex system with additional maintenance requirements	[25,26]
Façade-Integrated BIPVT	Façade	Energy savings, passive heating	Solar gain, cooling load reduction	35–55 %	11–15 %	9–17 %	$220–350/m^2, Payback: 7–9 years	10–25 % CO_2 reduction	Improves building insulation and reduces HVAC loads	Orientation-dependent; effectiveness varies with sun angle	[27]
Roof-Integrated BIPVT	Roof	Electrical & thermal efficiency, insulation benefits	PV module temperature, indoor cooling impact	45–65 %	12–18 %	10–20 %	$280–400/m^2, Payback: 6–8 years	20–35 % CO_2 reduction	Enhances roof insulation and eliminates need for separate PV mounting	Requires strong structural integration	[27,28]
Shading Device-Integrated BIPVT	Louvers, Blinds	Passive cooling, electricity generation	Shading efficiency, indoor comfort improvement	25–40 %	8–14 %	6–12 %	$180–280/m^2, Payback: 8–10 years	10–20 % CO_2 reduction	Enhances daylighting while generating electricity	Reduces direct solar gains but requires optimal orientation	[27,28]
PCM-Integrated BIPVT	Roof, Wall	Thermal storage, overheating mitigation	PCM phase transition efficiency, PV cooling	50–70 %	14–18 %	12–22 %	$350–500/m^2, Payback: 5–7 years	30–50 % CO_2 reduction	Stabilizes temperature fluctuations, improves night-time energy use	PCM degradation over time	[29]
Heat Pump-Integrated BIPVT	Roof, Façade	Heating & cooling efficiency	Heat pump COP, thermal comfort	55–80 %	13–20 %	15–25 %	$400–600/m^2, Payback: 4–6 years	35–55 % CO_2 reduction	Enhances heat recovery, ideal for cold climates	Expensive and requires active control systems	[30,31]
TEG-Integrated BIPVT	Roof, Wall	Waste heat recovery, additional power generation	Seebeck efficiency, temperature gradient stability	40–60 %	10–15 %	8–18 %	$300–450/m^2, Payback: 6–8 years	25–40 % CO_2 reduction	Converts heat loss into additional electricity	Low power output from TEGs limits efficiency gain	[32,33]

(*continued on next page*)

Table 2 (*continued*)

BIPVT Type	Building Integration	Studied Parameter	Assessment Parameters	Thermal Efficiency (%)	Electrical Efficiency (%)	Exergy Efficiency (%)	Cost Analysis ($/m^2 or Payback Period in Years)	Environmental Impact (CO_2 Reduction, Carbon Payback Time, etc.)	Major Findings	Remarks	Reference
Concentrated BIPVT	Roof	High-power PV efficiency	Optical concentration ratio, cooling requirements	60–85 %	18–24 %	15–28 %	$500–700/m^2, Payback: 4–5 years	40–60 % CO_2 reduction	Higher power output than conventional BIPVTs	Requires sophisticated cooling mechanisms	[9]
BIPVT with Thermal Wheels	Roof, Façade	Heat recovery, ventilation	Airflow rate, recovered heat percentage	50–75 %	10–18 %	12–20 %	$320–500/m^2, Payback: 6–8 years	30–45 % CO_2 reduction	Improves HVAC efficiency by transferring waste heat	Requires rotating components, increasing maintenance	[34]
BIPVT with Radiative Cooling & Advanced Insulation	Roof	Passive cooling, energy efficiency	Emissivity, heat dissipation rate	45–70 %	12–20 %	14–25 %	$280–450/m^2, Payback: 6–8 years	25–50 % CO_2 reduction	Reduces overheating using high-emissivity materials	Limited efficiency at night or in humid conditions	[35]
BIPVT with Solar Chimney System	Roof, Wall	Passive ventilation & cooling	Air velocity, cooling load reduction	40–65 %	10–16 %	10–22 %	$250–400/m^2, Payback: 6–7 years	20–40 % CO_2 reduction	Uses natural airflow for space cooling	Performance varies with wind speed and solar intensity	[36]
Proposed DSTS	Roof-Integrated Concrete Slab	Tri-generation: power, heating, and cooling	Thermal efficiency, PV output, passive cooling	65–82 %	16–22 %	18–30 %	$250–380/m^2, Payback: 5–7 years	50–70 % CO_2 reduction	Reduces indoor temperature by up to 6.2 °C, eliminates separate installations	Maximizes space efficiency, improves sustainability	Our Study

BIPVTs are among the most efficient and space-saving options for replacing conventional materials while generating electricity and heat. Unlike standalone PVTs, which require additional support, BIPVTs integrate seamlessly into buildings, thereby improving aesthetics, energy efficiency, and insulation. Their adaptability renders them ideal for sustainable construction. When integrated with other PVT systems, BIPVTs optimize thermal regulation and energy conversion. Pairing BIPVTs with water-based or hybrid PVTs enhances cooling and boosts the PV efficiency while providing hot water. BIPVT-TEG systems are particularly advantageous because they utilize waste heat to maintain stable temperature gradients, improving both power generation and passive cooling, which is an advantage of rooftop-mounted TEG-PVT systems. By embedding PV, TEGs, and heat exchangers within the building envelope, BIPVT-TEGs enhance energy harvesting, insulation, and overall sustainability. A detailed review of existing BIPVT systems is presented in Table 2.

Despite these advantages, PVT and BIPVT systems still require dedicated rooftop space, increasing the installation complexity and spatial constraints. While BIPVTs eliminate the need for separate mounting, many configurations involve additional layers, support structures, or external cooling mechanisms, which can limit the design flexibility and increase costs. Overcoming these challenges requires a fully integrated approach that minimizes structural modifications while maximizing multi-functionality. The proposed Domestic Solar Tri-Generation System (DSTS), a hybrid BIPVT-TEG system, addresses these issues by embedding PV panels, TEGs, and thermal absorbers directly within the concrete slab, thereby eliminating the need for separate rooftop installations. Unlike conventional roof-mounted PVT-TEG systems that demand additional space for PV panels, thermal collectors, and TEGs, our system optimizes the space efficiency while enhancing the structural integrity and energy performance. A key advantage of DSTS over conventional BIPVT systems is the use of the existing roof structure rather than the addition of extra layers or components. It does not require full roof coverage as it can be installed over specific sections, such as a single room's roof or an inclined rooftop structure, ensuring accessibility for other uses. This modular approach allows users to tailor installations based on their energy needs, space availability, and budget. Those seeking minimal installation can cover only part of their roof, reducing material costs and emissions, whereas larger installations can be deployed for higher energy generation. Thus, rather than compromising usability, DSTS transforms underutilized roof spaces into high-efficiency energy-generating surfaces.

As the DSTS is a hybrid BIPVT-TEG system, its advantages have been distinguished from conventional PVT and BIPVT technologies. However, as its core component is the solar concrete water heater, reviewing the reported building-integrated solar concrete water heater systems and their integration approaches is crucial for establishing its novelty. The following review presents the key advancements in solar concrete water heaters over the past three decades. In 2000, P.B.L. Chaurasia explored solar concrete collectors with embedded aluminium pipes that delivered water temperatures of 36–58 °C. Although this system is cost-effective and easy to install, the absence of glazing and insulation limits its efficiency [37]. Jay Burch et al. (2004) conducted simulations on unglazed solar concrete collectors, reporting that a 23 m^2 collector could fulfil 56 % of annual energy requirements, with performance varying based on system size [38]. In 2006, C. Dharuman et al. developed a building-integrated solar water heater with a 1.3 m^2 absorber, providing 100 L of hot water daily at 45°C-50 °C with 60 % efficiency during daylight hours. This system was 30 % cheaper than flat-plate collectors, but had minimal insulation benefits [39]. Luis Juanico et al. (2008) introduced a roof-integrated solar collector using a water pond for heating and cooling, offering an economical alternative to traditional roofing with enhanced thermal performance [40]. In 2010, Hazami et al. examined an integrated solar storage collector with copper pipes that met 80 % of a household's hot water needs in Tunisia [41]. Ji Jie et al. (2010) developed a dual-function collector for winter heating and summer water heating. The system achieved a water temperature of 42.5 °C and improved cooling efficiency by 2.05 % [42]. Later that year, Ji Jie et al. also tested a south-facing wall-mounted system that raised indoor temperatures to 24.7 °C compared to an ambient temperature of 4.8 °C, demonstrating a 4.2 °C stratification at room temperature. Selective absorbing coatings such as nickel-alumina improve thermal efficiency, correlating linearly with emissivity and indoor temperature [43]. In 2011, Sarachitti et al. integrated cement solar collectors into roofing, producing 40 L of hot water daily, reducing indoor temperatures by 2.3 °C, and achieving a payback period of 2.54 years [44]. M.S. Murthy et al. (2011) designed a low-cost water heater with copper tubing embedded in concrete slabs, achieving water temperatures of 60 °C and reducing costs by 75 % compared to conventional systems [45]. In 2014, Xiao Yu Sun et al. developed an all-ceramic solar collector with a high absorptance (0.93–0.97) and improved thermal performance [46]. V. Krishnavel et al. (2014) evaluated solar heaters using various materials, identifying aluminium pipes as the most effective and highlighting the potential of building roofs for dual heating and cooling [47]. Lin Su et al. (2015) modelled concrete ceiling radiant cooling panels, showing that reduced tube spacing enhanced cooling performance and that concrete's thermal inertia stabilized water temperatures [48]. Rui Li et al. (2015) tested balcony-mounted solar water heaters with 40 % collector efficiency, optimizing performance through lower tank volume-to-collector area ratios [49]. Hegarty et al. (2016, 2017) investigated concrete solar collectors and demonstrated that spacing, material conductivity, and flow rates affected performance. Their facade-integrated collectors supply up to 25 % of an individual's annual hot water needs, with solar fractions of 20 %, 24 %, and 30 % predicted for Helsinki, Dublin, and Sofia, respectively [50,51]. In 2017, Sable et al. developed an economical solar heater using dimpled copper tubes in concrete slabs, achieving water temperatures of 69 °C and short payback periods [52]. D. Prakash (2018) designed an insulated roof-integrated system producing 25 L per day with a 60 °C temperature rise by optimizing pipe diameter and insulation [53]. Celine Garnier et al. (2018) enhanced thermal stratification in integrated solar water heaters, boosting energy efficiency by 67 %–133 % while enabling seamless roof integration [54]. Adrian Pagsley et al. (2019) tested carbon nanotube-enhanced solar collectors, achieving 62 % efficiency and identifying nanotube integration as a future focus [55]. Sangram R. Patil et al. (2020) demonstrated a concrete solar collector with 42.75 % efficiency and 58 °C water output, emphasizing its durability and integration potential. Athikesavan Muthu Manokar et al. (2021) explored concrete collectors for cooling and heating, suggesting performance improvements through metal scraps and nanomaterials [56]. V.S. Chandrika et al. (2021) reported a thermal efficiency of 57 % for solar concrete water heaters in Tamil Nadu, proposing their use as replacements for conventional building envelopes [57]. Finally, Van Tram et al. (2023) investigated cooling pipe integration in reinforced concrete slabs,

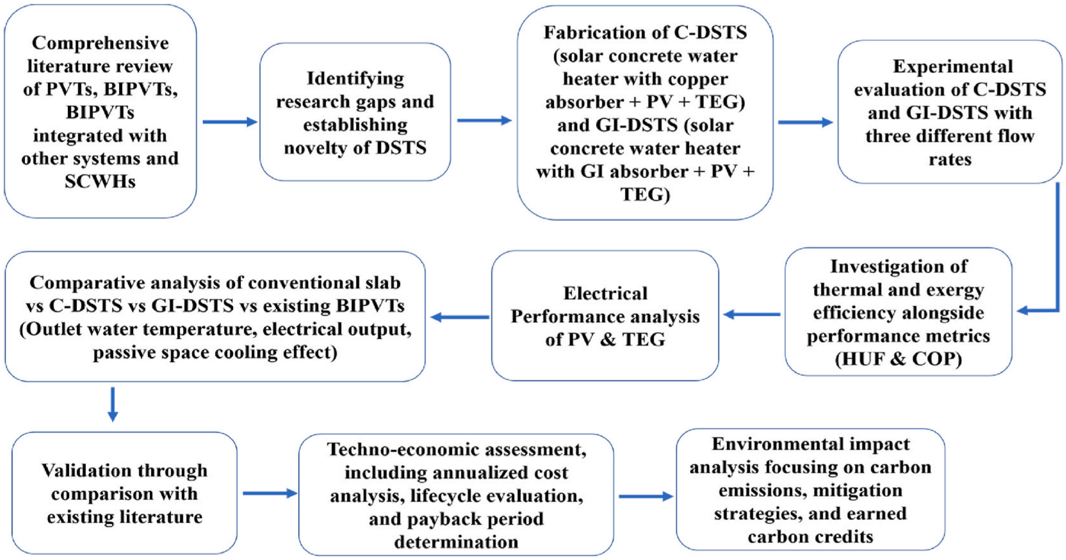

Fig. 1. Research methodology flow diagram.

optimizing flow rates to reduce thermal cracking and improving temperature regulation [58].

Over the past three decades, extensive research has focused on improving the efficiency of solar concrete water heaters using various building-integrated designs. Innovations include embedding aluminium or copper pipes within concrete and employing advanced structural integration techniques. However, most studies have focused on single-function applications, primarily solar water heating, with little emphasis on multifunctional features, such as PV power generation and thermoelectric cooling. Although conventional PVT systems have evolved into BIPVT configurations, challenges remain, including complex structural integration, seasonal performance fluctuations, additional material costs, and limited passive cooling benefits. Advanced BIPVT systems incorporating thermal wheels, radiative cooling, and phase-change materials offer better energy management, but require sophisticated auxiliary components, leading to higher costs and installation complexity. Additionally, TEG-integrated BIPVT systems have demonstrated potential for waste heat recovery; however, their efficiency is often constrained by unstable temperature gradients and inadequate passive cooling. To overcome these limitations, this study introduces a Domestic Solar Tri-Generation System (DSTS), a multifunctional solution integrating solar water heating, PV power generation, and thermoelectric cooling into a single concrete slab-based system that serves as a building roof. Unlike existing TEG-integrated BIPVT systems, where thermoelectric modules are attached as an additional layer, DSTS embeds TEGs beneath the concrete slab, optimizing the temperature gradient for continuous power generation while reducing indoor heat transfer through passive cooling mechanisms. The thermal mass of the concrete slab stabilizes the temperature fluctuations, optimizing both energy harvesting and indoor thermal comfort. Unlike existing TEG-integrated BIPVT systems, where thermoelectric modules are attached as an additional layer, the DSTS embeds TEGs beneath the concrete slab, optimizing the temperature gradient for continuous power generation while reducing indoor heat transfer. This dual functionality enhances both electricity generation and passive cooling, an advantage not found in traditional BIPVT-TEG configurations. Furthermore, integrating solar water heating within a concrete structure enhances the overall system efficiency, leading to better energy utilization and carbon footprint reduction. By combining PV, TEGs, and thermal storage into a single multifunctional unit, the DSTS surpasses conventional BIPVT solutions, offering improved durability, reduced maintenance, and enhanced thermal performance. The modular design ensures flexible deployment, making it a scalable and adaptable solution for sustainable building applications. This study positions the DSTS as a transformative advancement in building-integrated energy systems, paving the way for next-generation sustainable construction technologies.

The primary objectives of this study are as follows.

1. Design and construct copper-based (C-DSTS) and Galvanized Iron-based (GI-DSTS) Domestic Solar Tri-Generation Systems integrating solar concrete water heaters, photovoltaic (PV) panels, and thermoelectric generators (TEGs) for water heating, electricity generation, and space cooling.
2. To experimentally evaluate the combined performance of the DSTS components and their effectiveness in water heating, electricity production, and space cooling under various operational conditions.
3. To analyze the experimental results, quantifying the thermal and exergy efficiencies of the DSTS systems while evaluating the key performance indicators to validate the system efficiency and reliability.
4. To conduct a comprehensive techno-economic and environmental analysis, we assessed the financial feasibility and ecological benefits of DSTS systems compared to conventional alternatives.

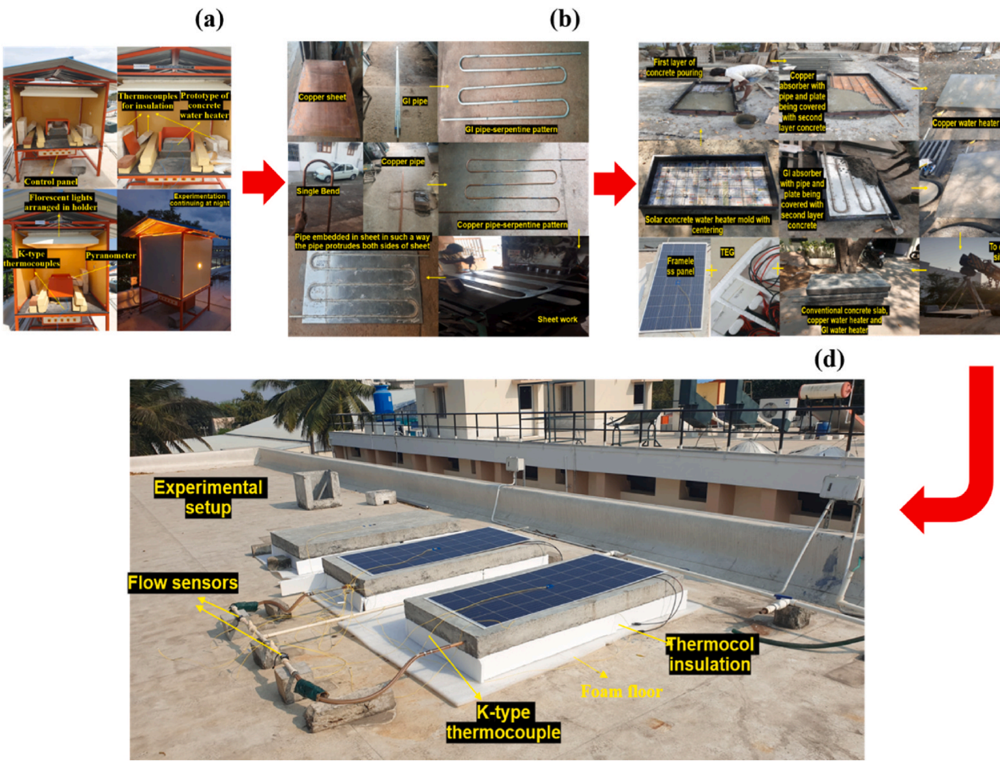

Fig. 2. (a) Prototype of a concrete slab in a thermal conductivity testing apparatus, **(b)** Fabrication of absorber for DSTS, **(c)** Solar concrete water heater fabrication pictorial, **(d)** DSTS experimental setup.

Table 3
Specifications of DSTS.

Specifications	Value
Concrete Water Heater	
Dimension	0.8 m (W) x 1.65 m (L) x 0.13 (H)
Grade	M20 grade (1:1.5:3)
Absorber	
Dimension	0.7 m (W) x 1.55 m (L) x 0.002 (T)
Absorber materials	Material 1: Copper (C) Material 2: Galvanized Iron (GI)
Absorber configuration	A serpentine-patterned pipe was embedded within the plate and designed such that the pipe extended outward from both sides of the plate.
Pipe details	Dimension: ID-0.0127 m/OD-0.0147 No. of pipes: 5 Length of pipes: 1.45 m-middle three/ 1.65 m-inlet and outlet
Plate dimension	0.7 m × 1.55 m x 0.002
PV Panel	
Dimension	0.7 m × 1.5 m x 0.0035 m
Type	Frameless-monocrystalline
Specs	Pmax-160 Wp/Vmp-18.8 V/Imp-8.52 A
TEG	
Dimension	0.15 m × 0.15 m × 0.0015 m
Specs at 25 °C	Qmax-50 W/Vmax-14.4 V/Imax-6.4 A
No. of TEGs/connection type	50/Parallel
General	
Tank Capacity	220 L
Pump	1.5 hp
Valve type	Ball valve
Insulation used	Polyurethane foam/thermocol
Connecting pipes	Chlorinated (C)-PVC

Table 4
Instrument specifications and uncertainty.

Instrument name	Measuring parameter	Range	Accuracy	External Uncertainty	Internal uncertainty
Pyranometer	Solar radiation	0–2000 W/m^2	±0.1 W/m^2	±0.14	–
K-type thermocouple	Temperature	− 200–650 °C	±0.01 °C	±0.14	±4.83
Flow sensor	Mass transfer rate	0–40 L/s	±0.01	±0.14	–
Multi-meter	Voltage & Current	200 mV-1000 V & 200 μA- 200 mA	±0.001	±0.014	–
Data acquisition system	Data storage	–	–	–	–

2. Methodology

2.1. Experimental set up

This study evaluated a Domestic Solar Tri-Generation System (DSTS) installed at the Vellore Institute of Technology in Vellore, India (12.9236° N, 79.1331° E). Fig. 1 depicts the steps followed in this project. A DSTS is an innovative setup designed to harness solar energy for three distinct purposes: electricity generation, water heating, and passive cooling. The core component of the system is a solar concrete water heater, which is integrated with a photovoltaic (PV) panel on its upper surface and thermoelectric generators (TEGs) on its underside to maximize the energy efficiency. Initially, the prototype slab was tested in a conductivity rig, as shown in Fig. 2 (a). Three experimental slabs were used: a conventional control slab, a slab with a copper-based absorber (C-DSTS), and a slab with a galvanized iron-based absorber (GI-DSTS). Both absorbers comprise a uniquely designed plate and pipe assembly, in which the pipe is centrally embedded within the plate, as depicted in Fig. 2 (b). This design, with a plate measuring 0.7 m in length, 1.55 m in width, and 0.002 m thickness, enhances heat transfer by aligning the pipe centrally along the absorber surface. The pipes had inner and outer diameters of 0.0127 m and 0.0147 m, respectively. Each slab includes five pipes: three central pipes, each 1.45 m long, and two end pipes, each 1.65 m long, for the fluid inlet and outlet. The absorber is embedded within an M20-grade concrete slab measuring 0.8 m × 1.65 m × 0.13 m as shown in Fig. 2 (c), enabling it to function as a solar water heater by transferring heat to the fluid circulating through the pipe network. Mounted atop each slab is a frameless monocrystalline PV panel, rated at 160 Wp and measuring 0.7 m × 1.5 m × 0.0035 m. Beneath the slab, 50 thermoelectric generators (TEGs) were arranged in a grid of ten rows with five TEGs per row to provide supplementary power and passive cooling functions. The experimental setup replicates household conditions by placing the concrete slabs on 0.15 m × 0.15 m square blocks over a polyurethane foam layer, creating an insulated indoor space beneath the DSTS roof. All four sides of the system were insulated with a thermocol to minimize heat loss. The water inlet connects to a 220 L overhead tank refilled by a 1.5 hp motor. The flow rates in the copper- and GI-based systems were regulated and monitored using ball valves and flow sensors positioned near the inlets, ensuring efficient fluid circulation and real-time parameter tracking. Fig. 2 (d) depicts a pictorial view of the experimental set up. Table 3 provides the detailed specifications of the DSTS.

2.2. Experiment

2.2.1. Instruments used and uncertainty analysis

Temperature readings were collected using K-type thermocouples, whereas flow rates and solar radiation were measured using flow sensors and a pyranometer, respectively. Voltage and current measurements were recorded using a multimeter. Data acquisition and storage were facilitated by using a Data Acquisition System (DAQ). Table 4 provides the details of each instrument, including specifications and associated uncertainty values. This study accounted for both external and internal sources of error to calculate the overall experimental uncertainty. The external uncertainty was determined to be 0.25 %, whereas the average internal uncertainty of the K-type thermocouples, which is essential for monitoring slab and panel temperatures, was calculated to be 4.83 %. Combining these factors resulted in an overall experimental uncertainty of 5.08 %. These values are consistent with those of previous research on solar concrete water heaters, confirming the reliability of the collected data for subsequent analyses [59]. The external uncertainty of the individual parameters can be calculated using Eq. (1).

$$External\ uncertainty = \sqrt{\left(\frac{\partial C}{\partial i1}\right)^2 u^2(i1) + \left(\frac{\partial C}{\partial i2}\right)^2 u^2(i2) + \ldots + \left(\frac{\partial C}{\partial in}\right)^2 u^2(in)} \tag{1}$$

where, C is the calculated parameter affected by independent parameters i1, i2 … in and $u(i1)$ $u(i2)$ and $u(in)$ are uncertainties of independent parameters.

The internal uncertainty of slab and panel temperature (sensitive parameter) can be attained using Eq. (2),

$$Internal\ uncertainty\ \% = \frac{\sqrt{sd'1^2 + sd'2^2 + \ldots + sd'n^2}}{mean\ value\ of\ total\ observations} \times 100 \tag{2}$$

where, sd' is the standard deviation of each step.

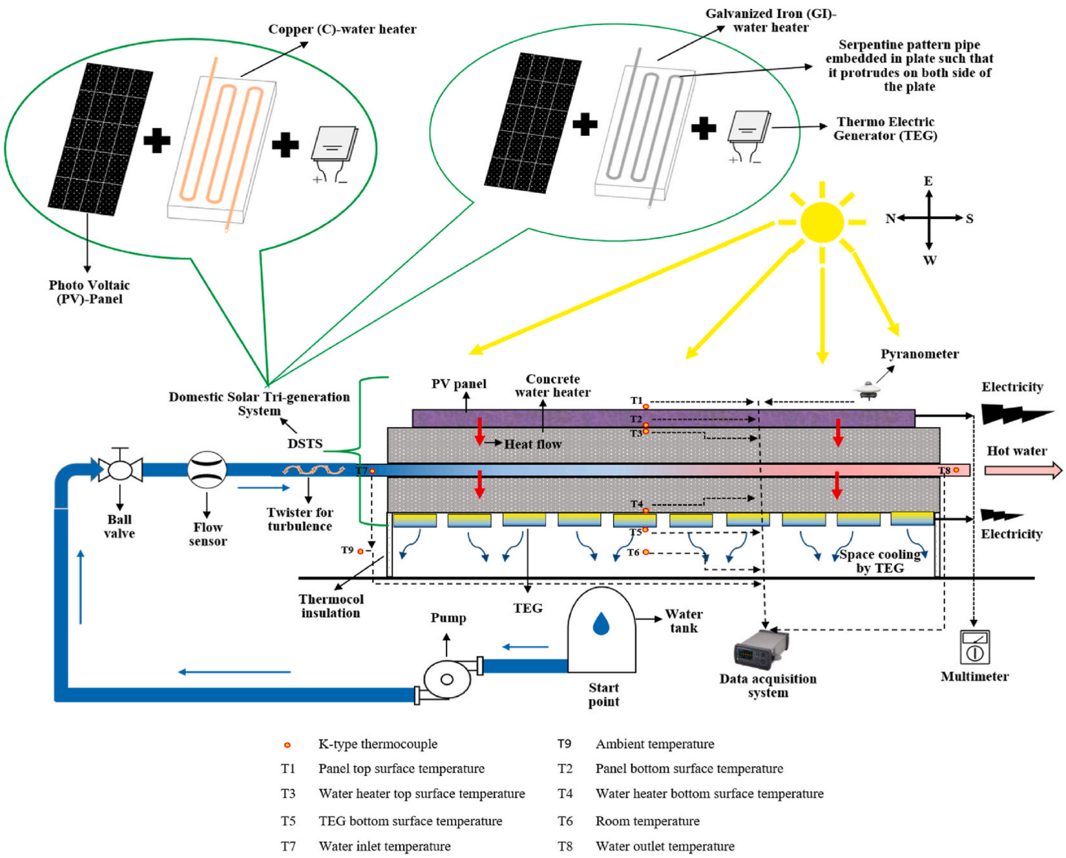

Fig. 3. Schematic of DSTS

2.2.2. *Experimental procedure*

The experimental procedure for a Domestic Solar Tri-Generation System (DSTS) was divided into three phases to evaluate its efficiency in power generation, hot water production, and passive cooling. Conducted in March 2024, data were recorded at 15-min intervals between 11 a.m. and 4 p.m. following guidelines from IS 3370-1 (2009) for concrete structures and IS 12976 (1990) for solar water heaters [60,61]. Initially, a prototype slab tested using a conductivity rig, as shown in Fig. 2 (a), yielded a thermal conductivity of 2.2 W/mK. The experimental setup included conventional, copper, and galvanized iron (GI) concrete slabs placed on concrete blocks insulated with a thermocol and polyurethane foam. K-type thermocouples, secured with M-Seal, monitored the slab surface temperatures, water inlet and outlet temperatures, and ambient indoor and outdoor conditions. PV panels and slabs were similarly instrumented, while water flow rates (0.12, 0.24, and 0.36 L/min) were managed using flow sensors. A Data Acquisition System (DAQ) logged all sensor data, and voltage and current readings were taken manually. The experiments demonstrated the capability of the DSTS to provide hot water, generate power, support cooling, and maximize solar energy usage in residential settings. Fig. 3 shows a schematic of the DSTS.

2.3. *Performance analysis*

The Reynolds and Prandtl number were obtained by Refs. [43,62],

$$R_e = \frac{\rho \times v \times Lp}{\mu} \tag{3}$$

$$Pr = \frac{\mu \times Cp}{kl} \tag{4}$$

where ρ denotes the fluid density, v denotes the fluid velocity, Lp denotes the pipe diameter, and μ denotes the dynamic viscosity. Cp is the specific heat capacity at constant pressure and kl is the thermal conductivity of the fluid.

The Nusselt number, the convective heat transfer coefficient-hc was calculated using the following equations [43,50,62],

$$Nu = 1.86 \times \left(\frac{R_e \times Pr}{\frac{L}{Lp}} \right)^{\frac{1}{3}} \tag{5}$$

$$hc = \frac{kl}{Lp} \times Nu \tag{6}$$

where, L is the pipe length

The Overall Heat Transfer Coefficient was calculated by Ref. [62],

$$\overline{\overline{U}} = \frac{1}{\frac{1}{hc1} + \frac{Lp}{kp} + \frac{Lconcrete}{kconcrete}} \tag{7}$$

where kp and $kconcrete$ are the thermal conductivities of the pipe and concrete, respectively, and $Lconcrete$ is the concrete length.

The useful heat gain by water was computed by Refs. [63,64],

$$\mathrm{Q}o = m \times Cp \times (tout - tin) \tag{8}$$

where m is the mass flow rate, $tout$ and tin are the outlet and inlet water temperatures, respectively.

Performance metrics were computed using the following equation:

The Heat Utilization Factor was obtained by Ref. [59],

$$HUF = \frac{m \times Cp \times (tout - tin)}{\dot{A} \times \overline{G} \times \widehat{T} \times \acute{F}} \tag{9}$$

where $\dot{A}$ is the surface area of the collector, $\overline{G}$ is the solar irradiance, $\widehat{T}$ is the solar exposure time, and $\acute{F}$ is the efficiency factor.

The heat loss from the collector was attained by Refs. [51,65],

$$\mathrm{Q}l = \dot{A} \times \overline{\overline{U}} \times (ts - ta) \tag{10}$$

where, ts is the collector surface temperature and ta is ambient temperature

The thermal efficiency of the system was obtained by Refs. [55,66],

$$\acute{\eta} T = \left(\frac{(\mathrm{Q}o - \mathrm{Q}l)}{\dot{A} \times \overline{G}} \right) \times 100\% \tag{11}$$

The efficiency factor was attained by Ref. [67],

$$\acute{F} = \frac{\mathrm{Q}o}{\dot{A} \times \overline{G}} \tag{12}$$

The Co-efficient Of Performance of the system was obtained by Ref. [64],

$$cop = \frac{\mathrm{Q}o}{\mathrm{Q}o + Ppump} \tag{13}$$

The exergy efficiency was calculated using the following equations [67,68],

$$Eo = m \times Cp \times (tout - ta) \times \left(1 - \frac{ta}{ts}\right) \tag{14}$$

$$Ei = \dot{A} \times \overline{G} \times \left(1 - \frac{ta}{ts}\right) \tag{15}$$

$$\acute{\eta} E = \frac{Eo}{Ei} \tag{16}$$

The efficiency of the photovoltaic panels was attained by Ref. [69],

$$\acute{\eta} P = \left(\frac{\dot{V} \times \dot{I}}{\dot{A} \times \overline{G}} \right) \times 100\% \tag{17}$$

where, $\dot{V}$ and $\dot{I}$ are output voltage and current from the panel.

2.4. Economic analysis

2.4.1. Annualized cost analysis

The Annualized Life Cycle Cost of DSTS ($ALCC_{DSTS}$) is computed by the following equations [70,71]

$$ALCC_{DSTS} = ALCC + \left[\mathscr{C}_{EWH} + \mathscr{C}_{cooling}\right] - \mathscr{C}_{PV-TEG} \tag{18}$$

where $ALCC$ is the annualized life cycle cost of C and GI-DSTS, $\mathscr{C}_{EWH}$ is the annual energy cost avoided by replacing the electric water heater, $\mathscr{C}_{cooling}$ is the annual energy cost avoided by replacing the electric cooling systems (fans/ACs), $\mathscr{C}_{PV-TEG}$ and is the annual savings from the electricity generated by the PV and TEG.

The $ALCC$ can be attained by following equation,

$$ALCC = LCC \times \mathscr{C}_{rf} \tag{19}$$

The Life Cycle Cost (LCC) is attained by,

$$LCC = \mathscr{C}_i + \sum_{t=1}^{n}\left[\left(\mathscr{C}_{o,t} + \mathscr{C}_{m,t} + \mathscr{C}_{r,t}\right) \times Df_t\right] - \mathscr{C}_s \times Df_n \tag{20}$$

where, $\mathscr{C}_i$ is initial investment cost, $\mathscr{C}_{o,t}$ is operation cost, $\mathscr{C}_{m,t}$ is maintenance cost, $\mathscr{C}_{r,t}$ is replacement cost, $\mathscr{C}_s$ is the salvage value at the end of system life time in years (n), Df_t and Df_n are the discount factor. where $\mathscr{C}_i$ is the initial investment cost, $\mathscr{C}_{o,t}$ is the operation cost, $\mathscr{C}_{m,t}$ is the maintenance cost, $\mathscr{C}_{r,t}$ is the replacement cost, $\mathscr{C}_s$ is the salvage value at the end of the system lifetime in years (n), Df_t and Df_n are discount factors.

The Capital recovery factor ($\mathscr{C}_{rf}$) is obtained by,

$$\mathscr{C}_{rf} = \frac{r \times (1+r)^n}{(1+r)^n - 1} \tag{21}$$

where, r is the real interest rate.

2.4.2. Life cycle savings estimation

The inflation adjusted annual savings ($\mathscr{A}_i$) is computed by Ref. [72],

$$\mathscr{A}_i = \mathscr{A}_{initial} \times (1+i)^n \tag{22}$$

where, $\mathscr{A}_{initial}$ is the initial annual saving, i is the inflation rate.

The degradation adjusted annual savings ($\mathscr{A}_d$) is computed by,

$$\mathscr{A}_d = \mathscr{A}_{initial} \times (1-d)^n \tag{23}$$

where, d is the degradation rate.

The cumulative savings ($\mathscr{A}_c$) is obtained by,

$$\mathscr{A}_c = \sum_{t=1}^{n} \mathscr{A}_i + \mathscr{A}_d \tag{24}$$

The present value of savings ($\mathscr{PV}$) is obtained by,

$$\mathscr{PV} = \frac{\mathscr{A}_c}{(1+r)^n} \tag{25}$$

2.4.3. Payback period

$$\mathscr{P} = \frac{\mathscr{C}_i}{\mathscr{A}_{initial}} \tag{26}$$

2.5. Environmental analysis

The CO_2 emissions of DSTS are determined using equations that incorporates 20 % appliance losses and 40 % transmission and distribution losses [73],

$$Emission\ of\ CO_2\left(\frac{kg}{year}\right) = \frac{\hat{E} \times 2.04}{n} \tag{27}$$

where, Ê is the total embodied energy of DSTS.

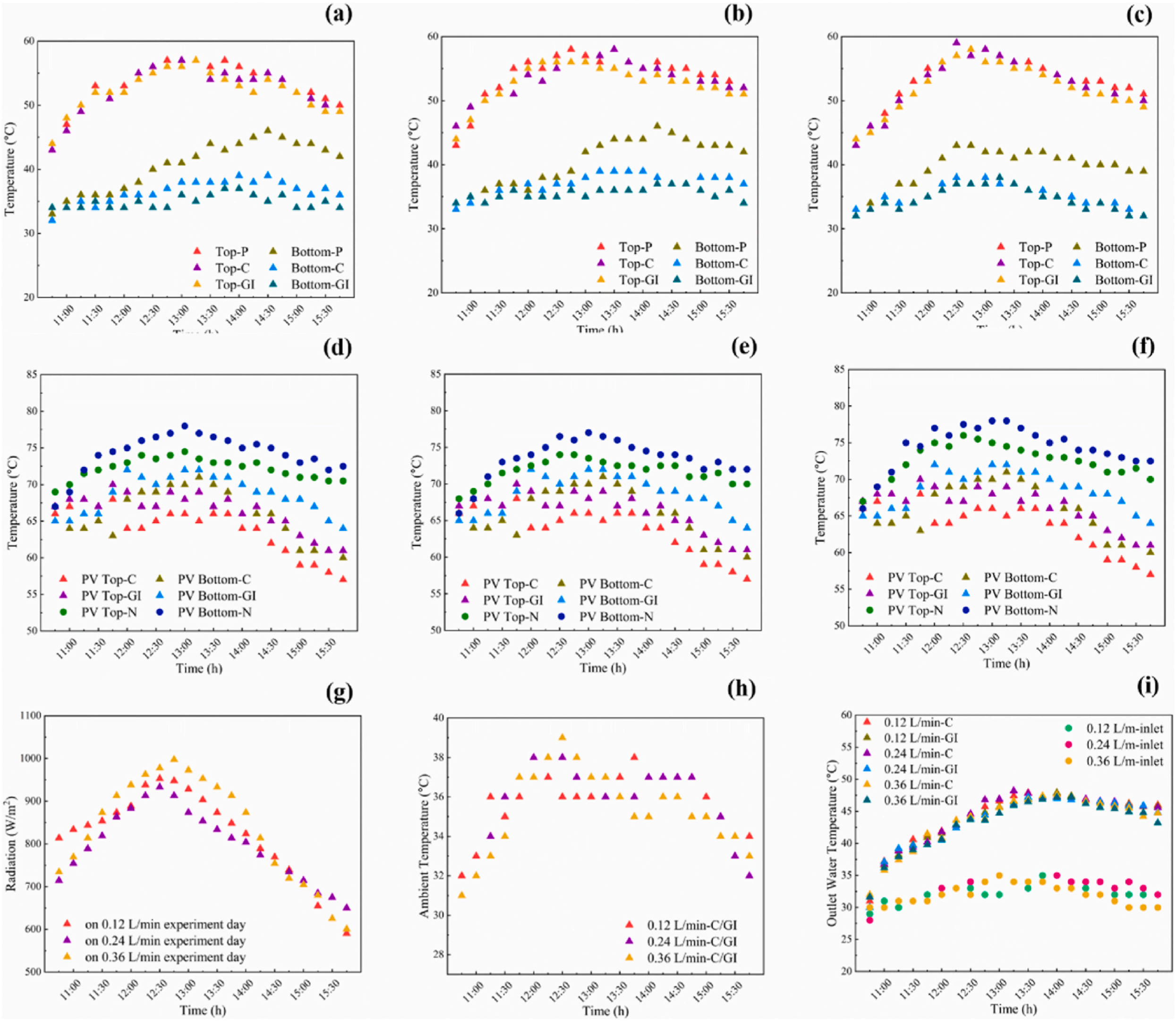

Fig. 4. Concrete's top and bottom temp. at: **(a)** 0.12 L/min, **(b)** 0.24 L/min, **(c)** 0.36 L/min; PV panel's top and bottom temp. at: **(d)** 0.12 L/min, **(e)** 0.24 L/min, **(f)** 0.36 L/min; **(g)** Radiation, **(h)** Ambient temperature and **(i)** Outlet water temperature.

$$Mitigation\ of\ CO_2\left(\frac{kg}{year}\right) = \hat{E}T \times 2.04 \tag{28}$$

where, $\hat{E}T$ is DSTS's annual thermal energy output that is obtaoned by,

$$\hat{E}T = DT \times ns \tag{29}$$

where, ns represents the sunny days annually and DT represent thermal energy output of DSTS daily, computed by,

$$DT = Qh + Qc \tag{30}$$

where Qh and Qc are the thermal energies used for water heating and space cooling, respectively.

$$Net\ CO_2\ mitigation\ (Lifetime) = \left[(\hat{E}T \times n) - \hat{E} \times 2.04 \times 10^{-3}\right] \tag{31}$$

$$CO_2\ \ credit\ earned = Net\ CO_2\ mitigation \times Cost\ of\ CO_2\ mitigation\ per\ ton \tag{32}$$

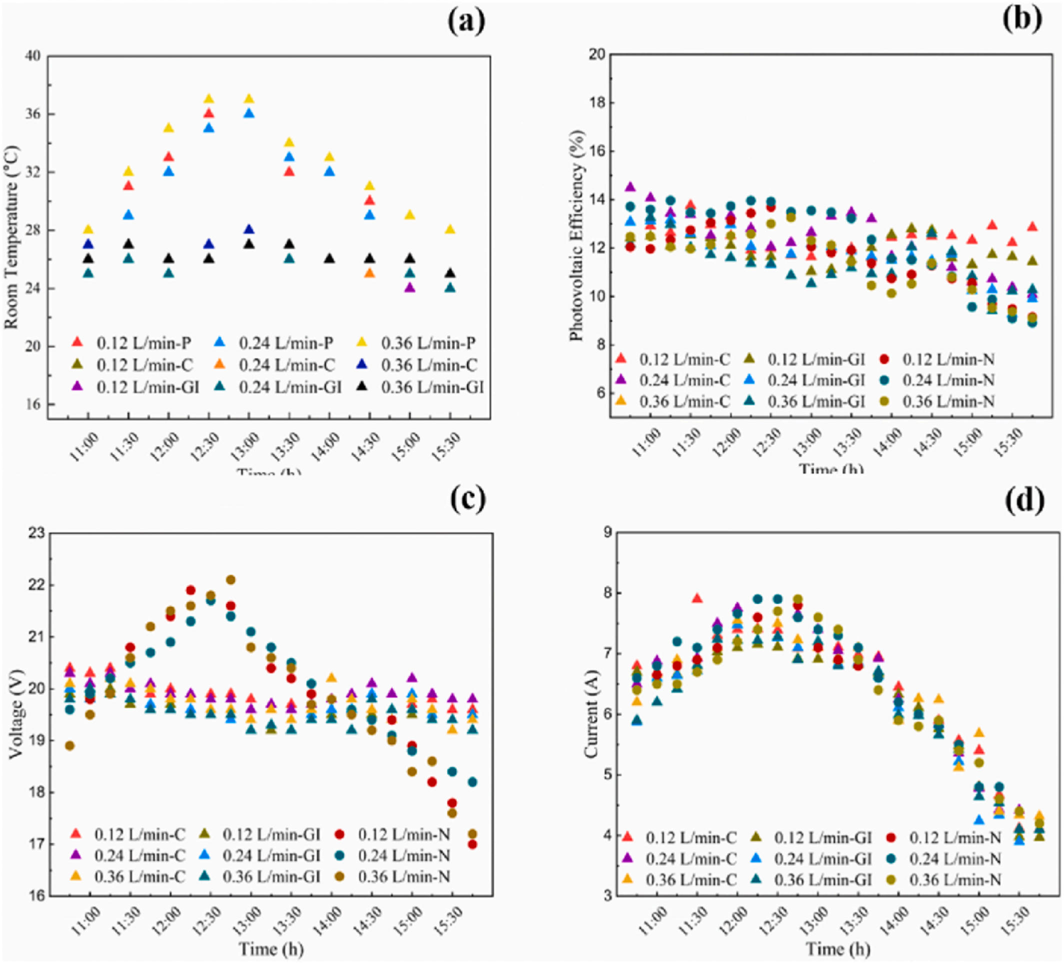

Fig. 5. (a) Room temperature under three slabs, **(b)** Photovoltaic efficiency, **(c)** Voltage and **(d)** Current.

3. Results and discussions

3.1. Experimental results

Tests were conducted on a standard concrete slab, C-DSTS, and GI-DSTS at flow rates of 0.12, 0.24, and 0.36 L/min over consecutive days to evaluate their thermal heating and cooling performance. The top and bottom surface temperatures of all three slabs, along with the PV panel temperatures at different flow rates, are shown in Fig. 4(a)–(f). For the conventional concrete slab, the top surface temperature peaked between 52 and 59 °C, with an average range of 48.57–53.76 °C, while the bottom surface recorded maximum temperatures of 43–46 °C, averaging 37.28–40.57 °C. The resulting temperature difference between the top and bottom surfaces ranged from 9.29 to 13.66 °C, indicating moderate heat-retention properties. In contrast, the C-DSTS slab exhibited higher thermal efficiency, with top surface temperatures ranging from 55 to 59 °C, averaging 49.09–53.28 °C. The bottom surface temperature peaked at 39 °C, with an average of 33.61–37 °C, leading to a temperature difference of 15.19 17.38 °C. The GI-DSTS slab, on the other hand, recorded top surface temperatures of 52 °C–58 °C, with an average of 52–52.71 °C, while the bottom surface peaked at 38 °C, averaging 34.61–35.52 °C, resulting in a temperature difference of up to 17.39 °C.

The PV panel temperatures varied across both DSTS systems. The C-DSTS PV panel recorded a top surface peak of 70 °C, averaging 65.33 °C, while the bottom surface peaked at 72 °C, averaging 67.85–68.38 °C. The GI-DSTS PV panel had a top surface peak of 72 °C, averaging 65.23–67 °C, and a bottom surface peak of 74 °C, averaging 67.85–69.76 °C. The solar irradiance levels, as shown in Fig. 4(g), ranged from 778.44 to 834.45 W/m^2, with peak values between 903.22 and 997.51 W/m^2. Meanwhile, ambient temperatures, as illustrated in Fig. 4(h), varied between 35.33 and 36.14 °C, reaching maximum values of 39–43.6 °C.

A comparison of the inlet and outlet water temperatures for both the C-DSTS and GI-DSTS at various flow rates is presented in Fig. 4(i). The inlet water temperature ranged between 30.66 and 32.90 °C, while the C-DSTS outlet water temperature peaked at 48.2 °C, averaging between 39.46 and 43.95 °C, reflecting a temperature increase of 9.97–11.42 °C. The GI-DSTS outlet water temperature peaked between 47.6 and 48.2 °C, averaging 39.49–43.95 °C, with a temperature rise of 9.45–11.05 °C.

Table 5
Maximum error values for each metric in validation.

Metric	Max Error (C-DSTS)	Max Error (GI-DSTS)
Outlet Water Temperature (°C)	14.4	14.80
Bottom Roof Temperature (°C)	5	4
Room Temperature (°C)	1	1
Heat Utilization Factor (HUF)	0.08	0.12
Coefficient of Performance (COP)	0.72	0.6
Thermal Efficiency (%)	12.02	11.22
Exergy Efficiency (%)	8.08	7.18

The results indicate that the C-DSTS consistently achieved higher outlet water temperatures, surpassing the GI-DSTS by 0.39–0.82 °C. Additionally, both DSTS systems demonstrated better thermal performance than similar systems, as shown in Fig. 7(a), reinforcing their efficiency in water heating applications [42,52,57]. Compared to the conventional slab, the C-DSTS reduced the bottom surface temperatures by 3.57–4.71 °C, while the GI-DSTS showed additional reductions of 0.48–2.23 °C when compared to the C-DSTS. An analysis of the DSTS bottom roof temperatures, compared with previous studies (Fig. 7(b)), demonstrated notable reductions, contributing to enhanced indoor thermal comfort through passive cooling, without any mechanical components or external energy input [44,53].

The cold-side temperatures of the TEGs were measured every 30 min. For C-DSTS, the bottom TEG temperature averaged 23.4 °C, 23.3 °C, and 23.9 °C at flow rates of 0.12, 0.24, and 0.36 L/min, respectively. The GI-DSTS exhibited similar values, with 23.9 °C, 24.1 °C, and 23.9 °C, respectively. The TEGs in both the systems, connected in parallel, produced an average voltage of 0.685 V and a current of 50 A.

Room temperatures were recorded every 30 min, and the results are shown in Fig. 5(a). At 0.12 L/min, the room temperatures for the conventional slab, C-DSTS, and GI-DSTS were 31.7 °C, 25.69 °C, and 25.9 °C, respectively. At 0.24 L/min, the values were 31 °C, 25.9 °C, and 25.6 °C, while at 0.36 L/min, they were 32.4 °C, 26.5 °C, and 26.2 °C. A comparison of the space cooling performance, shown in Fig. 7(c), highlights the DSTS's significant ability to lower indoor temperatures and improve thermal comfort [42–44]. Finally, Fig. 5(b), (c), and (d) present the photovoltaic efficiency, voltage, and current output of the system, respectively, confirming the effectiveness of the DSTS in integrating power generation, space cooling, and water heating functionalities.

3.2. *Performance analysis*

The study assessed the Heat Utilization Factor (HUF), Coefficient of Performance (COP), thermal efficiency, and exergy efficiency of DSTSs across different flow rates (0.12, 0.24, and 0.36 L/min), with optimal performance observed at 0.36 L/min. C-DSTS consistently exhibited superior results across all parameters owing to its high thermal conductivity, which facilitated more efficient heat transfer. However, the GI-DSTS demonstrated better heat retention and redistribution, narrowing the performance gap when integrated with PV panels and thermoelectric generators (TEGs). As the flow rate increased, the average HUF also improved, reaching 0.093, 0.177, and 0.269 for the C-DSTS and 0.090, 0.170, and 0.260 for the GI-DSTS. Similarly, the COP values increased with higher flow rates, averaging 1.203, 2.265, and 3.513 for the C-DSTS, while the GI-DSTS achieved COP values of 1.161, 2.162, and 3.393, respectively. The HUF and COP of the proposed system surpassed similar configurations reported in the literature, as depicted in Fig. 7 (d) and (e) [62,64,74,75], and, demonstrating its efficiency in energy utilization. These findings highlight the scalability and practicality of DSTS.

The thermal efficiency peaked at 24.75 % for the C-DSTS and 24.20 % for the GI-DSTS at 0.36 L/min, both exceeding the performance of benchmark systems reported in the literature, as illustrated in Fig. 7(f) [49,55,66]. Exergy efficiency also improved with increasing flow rates, reaching a maximum of 37.05 % for C-DSTS and 32.30 % for GI-DSTS at 0.36 L/min. The system effectively minimized energy losses, achieving superior energy conversion quality through passive mechanisms when compared to other designs, as shown in Fig. 7(g) [67,68], particularly the thermoelectric generators which utilize existing temperature gradients without requiring external power for operation.

The validation results demonstrate that the maximum error values (Table 5) between the proposed C-DSTS and GI-DSTS systems and the reference studies are within an acceptable range, with deviations primarily attributed to variations in experimental conditions, material properties, and system configurations. The highest observed errors were in the thermal efficiency (12.02 %) and exergy efficiency (8.08 %), which remained within reasonable experimental limits. These discrepancies can be explained by differences in measurement methodologies, environmental factors, and inherent uncertainties in thermophysical properties, reinforcing the reliability of the performance of the proposed system.

Although the DSTSs' PV panels were installed without inclination, their electrical efficiency remained comparable to that of a conventional PV panel tested under similar conditions. This was primarily due to the cooling effect provided by the water heater and TEGs. The PV temperature trends depicted in Fig. 4(d), (e), and (f) indicate that the conventional panel experienced significantly higher top and bottom surface temperatures than the panels integrated into the DSTSs, highlighting the effectiveness of the concrete water heater and TEGs in reducing the thermal stress.

The electrical efficiency analysis presented in Fig. 5(b) shows that at 0.12 L/min, the conventional panel, C-DSTS panel, and GI-DSTS panel achieved average efficiencies of 11.69 %, 12.39 %, and 11.93 %, respectively. At 0.24 L/min, the efficiencies were recorded as 12.31 %, 12.33 %, and 11.71 %, while at 0.36 L/min, they were 11.42 %, 11.38 %, and 11.38 %. The system effectively

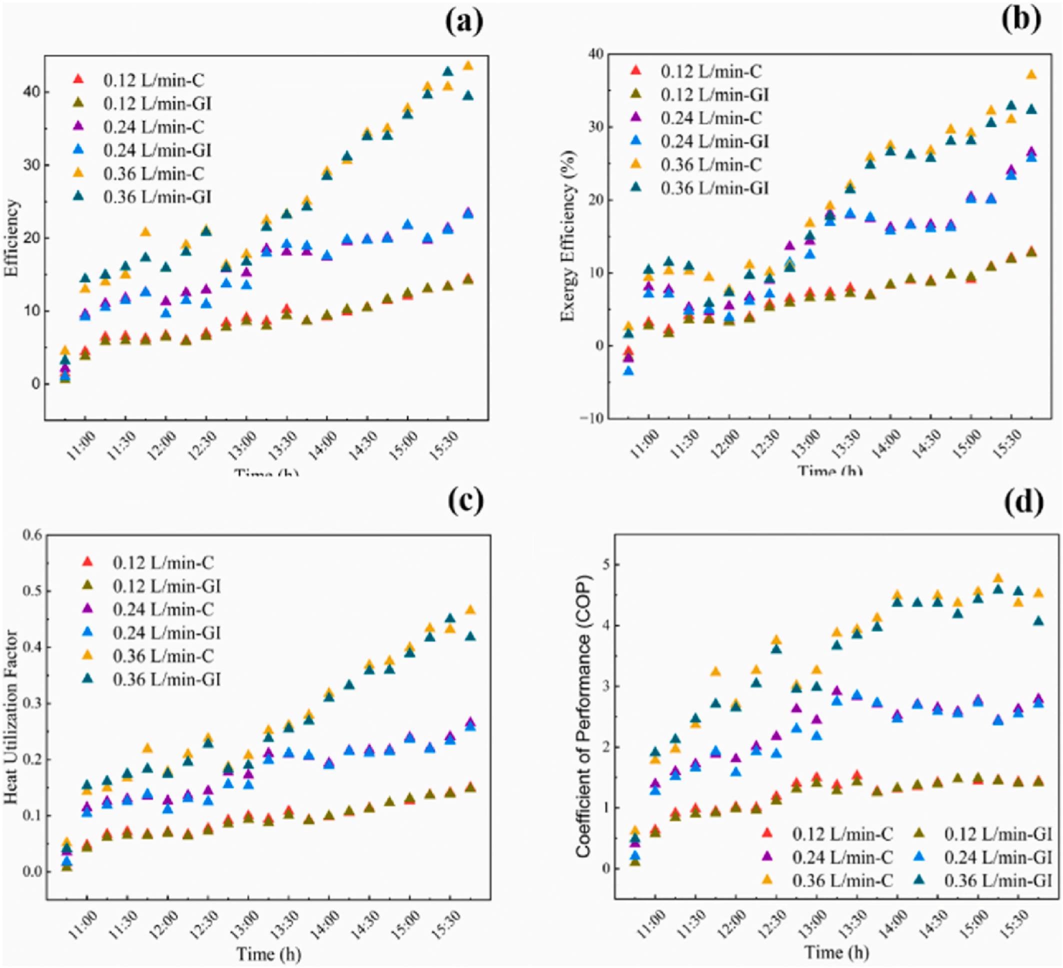

Fig. 6. (a) Thermal efficiency, **(b)** Exergy efficiency, **(c)** Heat utilization factor and **(d)** Coefficient of performance.

lowered indoor temperatures from 31.4 °C in the conventional setup to as low as 25.69 °C, significantly enhancing thermal comfort. The overall efficiency of the DSTS was 46.4 %, with TEGs contributing an average efficiency of 11 %. The comparative evaluation confirmed the multifunctional capabilities of the DSTS in generating electricity, heating water, cooling indoor spaces, and regulating the PV panel temperatures. The seamless integration of PV, TEGs, and solar concrete water heaters in the DSTS provides a sustainable and efficient energy solution, delivering significant environmental and operational benefits while maintaining competitive performance across all metrics.

3.3. Building implementation

The developed Domestic Solar Tri-Generation System (DSTS) was designed with a strong focus on building integration. The reinforced concrete slab used in the experiments replicates a typical residential roof, functioning both as a structural element and an energy-harvesting surface. This dual role minimizes space requirements and is especially suitable for urban buildings with limited roof area. The frameless PV module mounted above the slab ensured on-site electricity generation while maintaining durability and aesthetics. Embedded serpentine copper and GI pipes delivered hot water for household use, with results (Fig. 6(a)) showing that 108 L of water was heated to 50 °C within ~5 h of solar exposure—meeting daily residential demand. In addition, thermoelectric modules placed between the PV and slab enabled passive space cooling, lowering indoor air temperature by up to 6.2 °C (Fig. 6(c)). This directly reduces cooling load and energy demand. The concrete slab further enhances compatibility with standard construction while leveraging its thermal mass for heat retention and release. The mock-roof test arrangement, incorporating insulation, simulated real rooftop conditions and validated system applicability for residential use. These findings confirm that the DSTS can be effectively implemented in both new buildings and retrofits, offering a multifunctional and sustainable alternative to conventional energy systems.

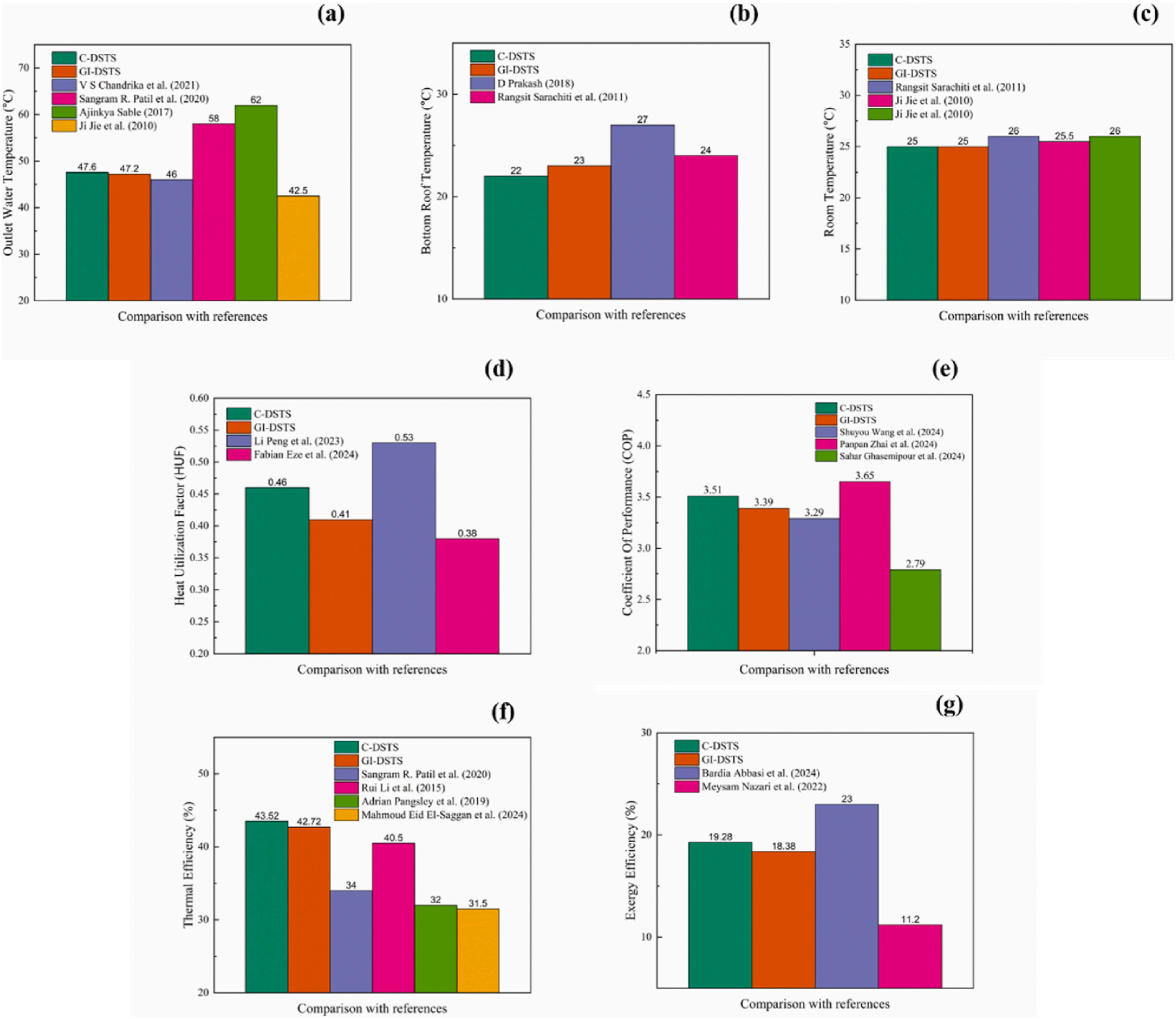

Fig. 7. Comparison of DSTS with previous works available in literature: **(a)** Outlet water temperature, **(b)** Bottom roof temperature, **(c)** Room temperature, **(d)** Heat utilization factor, **(e)** Coefficient of performance, **(f)** Thermal efficiency and **(d)** Exergy efficiency.

Table 6
Economic and cost Parameters of DSTSs [59].

Material and labour Cost	C-DSTS-1321.06 USD GI-DSTS-855.06 USD
Maintenance Cost	5 % of initial investment
Replacement Cost	2 % of initial investment
Salvage value	5 % of initial investment
Rate of interest	8 %
Rate of inflation	5 %
Rate of real interest	3 %
The price of electricity (USD/kWh)	0.048
Life span of DSTS	25 years
Life span of conventional systems	10 years

4. Economic analysis

An economic study evaluated the profit potential of the DSTS and compared it with that of conventional electric systems. The economic analysis of the DSTS is presented using three methods: annualized cost analysis, life cycle evaluation, and payback period.

Table 7
Annual savings of GSTD assessed for each year throughout the life and its real-time worth.

Year	Basic Annual Saving (USD)	Inflation-Adjusted Saving (USD)	Degradation-Adjusted Saving (USD)	Cumulative Saving (USD)	Present Value (USD)
1	161.76	161.76	161.76	161.76	157.04
2	160.15	168.09	166.41	328.17	157.99
3	158.54	176.49	171.08	499.25	158.95
4	156.96	185.32	175.79	675.04	159.93
5	155.39	194.58	180.53	855.57	160.93
6	153.83	204.31	185.31	1040.88	161.94
7	152.30	214.53	190.12	1231.00	162.96
8	150.77	225.25	194.96	1425.96	164.00
9	149.27	236.51	199.84	1625.80	165.06
10	147.77	248.34	204.76	1830.55	166.13
11	146.30	260.76	209.71	2040.26	167.21
12	144.83	273.80	214.69	2254.95	168.32
13	143.38	287.49	219.72	2474.67	169.44
14	141.95	301.86	224.78	2699.45	170.58
15	140.53	316.95	229.88	2929.34	171.73
16	139.13	332.82	235.02	3164.36	172.90
17	137.73	349.51	240.21	3404.57	174.09
18	136.36	367.07	245.43	3649.99	175.30
19	134.99	385.55	250.69	3900.68	176.52
20	133.64	405.02	255.99	4156.68	177.77
21	132.31	425.53	261.34	4418.02	179.03
22	130.98	447.15	266.73	4684.75	180.31
23	129.67	469.94	272.17	4956.92	181.61
24	128.38	492.80	277.65	5234.57	182.92
25	127.09	516.98	283.18	5517.75	184.26

The economic and cost parameters of the DSTS are summarized in Table 6.

4.1. Comparison of DSTSs and conventional electric systems by annualized cost analysis

The annualized cost method is an effective tool for comparing systems with different lifespans that perform the same functions. In this analysis, the annualized costs were calculated using equations (18)–(21). Solar technologies in India typically operate for approximately 290 d per year. For this study, it was assumed that both Domestic Solar Tri-Generation Systems (DSTSs) would operate for 245 days annually, with hot water production, power generation, and space cooling considered for 5 h per day, matching the experimental duration. Within this timeframe, both DSTSs can annually produce 26,460 L of hot water, generate 198 kWh of electricity, and lower room temperatures by up to 6.2 °C during their operational hours. The fabrication costs for the C-DSTS and GI-DSTS are 1321.12 USD and 855.10 USD, respectively (based on an exchange rate of 1 USD = ₹85.03), excluding maintenance costs, which are 5 % of the initial investment. The difference in the initial investment arises from the higher cost of copper compared to that of galvanized iron. The DSTSs were compared with conventional systems in terms of their ability to deliver equivalent functions. Conventional systems include electric water heaters and a combination of fans and air conditioners, all of which are assumed to operate with 80 % efficiency. The annual energy cost of operating a conventional electric water heater is 40.46 USD, and the cost of cooling systems to achieve the same cooling effects as the DSTSs is 109.95 USD. By contrast, both DSTSs have zero annual energy costs because they rely solely on solar energy. The annualized costs of the C-DSTS, GI-DSTS, and conventional systems are 167.26 USD, 108.25 USD, and 286.30 USD, respectively. Based on these costs, the cumulative annual cost for heating 1 L of water by 1 °C and reducing room temperature by 1 °C is 13.76 USD for the C-DSTS, 18.13 USD for the GI-DSTS, and 13.29 USD for the conventional systems. Conventional systems directly rely on electricity for heating and cooling, making their energy costs straightforward to calculate. However, their shorter lifespans (10 years) contrast with the 25-year lifespan of DSTSs, which includes embedded construction costs. The higher apparent costs stem from the cost distribution over their lifespan and partial equivalence of their cooling functions. While DSTSs' benefits are calculated for a 5-h daily operation, actual usage often extends beyond this, providing greater advantages. The multi-functionality, environmental benefits, and reliance on renewable energy of the DSTSs justify their costs.

4.2. Life cycle savings

Life-cycle savings estimates the savings that an asset will generate over its lifetime. This method involves determining the daily savings by computing the costs spent at each temperature. Despite the variation in the initial investment costs between C-DSTS and GI-DSTS, their energy cost savings are nearly identical owing to comparable performance outputs. Therefore, energy cost savings were commonly calculated based on average performance results to ensure consistency. The daily savings of the DSTSs are 0.17 USD for water heating and 0.45 USD for space cooling. Table 7 using equations (24)–(27) illustrates the annual and cumulative financial benefits of a Domestic Solar Trigeneration System (DSTS) over a 25-year lifespan. Starting with a basic annual saving of 161.58 USD in year 1, the system's financial advantages gradually adjust to account for inflation, system degradation, and the time value of money. By

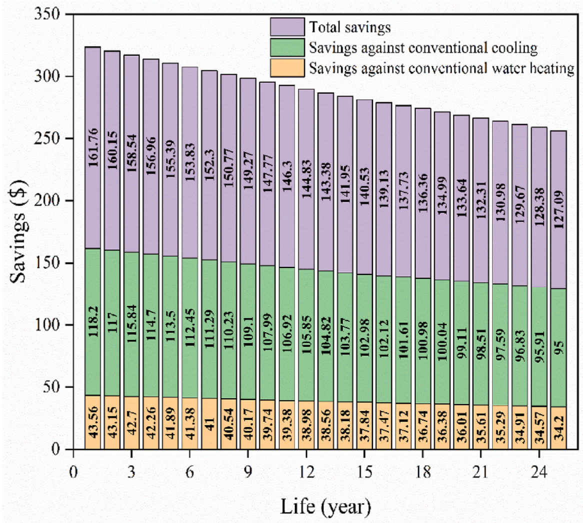

Fig. 8. Year-on-year cost saving.

Table 8
Embodied energy of materials used in DSTSs fabrication [59,79].

S. No.	Items	Coefficient of embodied energy (kWh/kg)	Amount (kg)	Total embodied energy (kWh)
Absorber plate				
1.	Copper plate	29.17	19.43	566.77
2.	GI plate	10.56	17.04	179.94
3.	Copper pipe	29.17	5.025	146.56
4.	GI Pipe	10.56	4.68	49.4
5.	Copper welding tool	29.17	0.2	5.83
6.	GI welding tool	10.56	0.2	2.11
Concrete				
7.	Cement	1.556	44.93	69.91
8.	Fine aggregate	0.02778	74.88	2.08
8.	Coarse aggregate	0.02778	140.4	3.9
9.	Centring rod	6.945	10.39	72.15
Miscellaneous				
10.	Panel	19.20	11.2	215.1
11.	Thermoelectric generator	11.11	50 Nos × 1.672	555.5
12.	Ball valve	0.502	1 No	7.39
13.	Connecting pipes (PVC)	–	6.1m	13.64
		Total	**C-DSTS =**	**1658.83 kWh**
			GI-DSTS =	**1171.12 kWh**

year 25, the cumulative saving reaches 55113.57 USD, with the present value of these savings calculated at 43265.77 USD. Fig. 8 shows the year-wise cost savings of the DSTS. This analysis highlights the ability of the DSTS to deliver significant long-term savings, especially when compared with conventional systems, demonstrating its economic and operational efficiency.

4.3. Payback period

The payback period denotes the time it takes for an investor to recoup the full initial investment in a project through savings. With a capital investment of 1321.06 USD and 855.06 USD for C-DSTS and GI-DSTS respectively, the payback period for C-DSTS is 8.18 years and 5.29 years for GI-DSTS. Because of their short payback period and long operational lifespan, both DSTSs may be run at low operating costs throughout the vast majority of their lifecycle. DSTS's payback period of DSTS was compared to those of solar water heaters in the literature, and it was discovered that DSTS had a suitable payback period [73,76].

5. Environmental analysis

The assessment of CO_2 emissions is a critical aspect of evaluating the proposed DSTSs because it provides insight into their environmental impact. Environmental analysis was conducted for both the C-DSTS and GI-DSTS, with CO_2 emissions calculated based on their embodied energy, as shown in Table 8. Embodied energy is the total energy required to produce a product or service. The

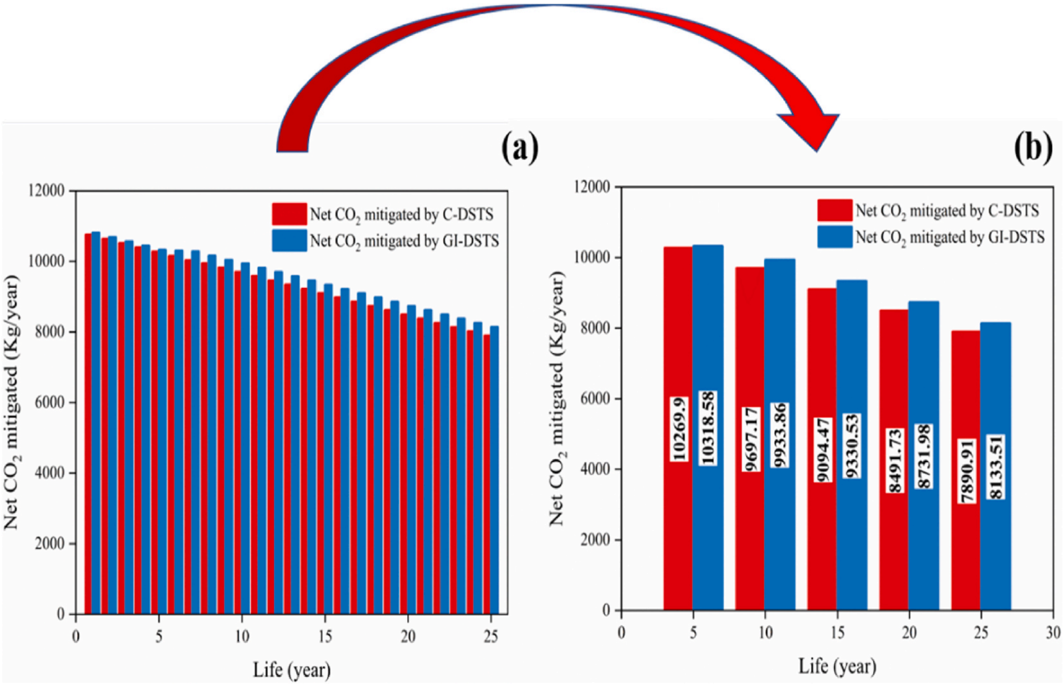

Fig. 9. (a) Net CO_2 mitigation for 25 years, **(b)** Every 5-year net CO_2 mitigation.

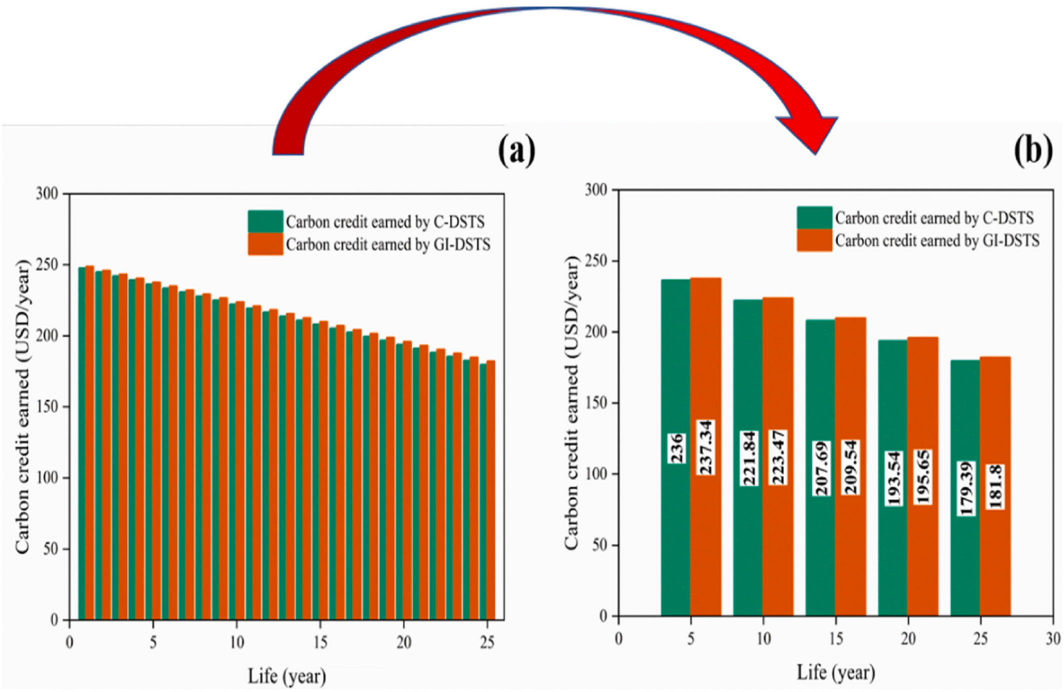

Fig. 10. (a) Carbon credit earned for 25 years**, (b)** Every 5-year carbon credit.

production of C-DSTS and GI-DSTS utilized approximately 1658.83 kWh and 1171.12 kWh of embodied energy, respectively. Over their lifetime, the CO_2 emissions for C-DSTS were calculated as 85,935 kg, while GI-DSTS emissions amounted to 59,722.25 kg. The annual CO_2 mitigation for both systems was directly linked to the energy generated. The developed C-DSTS and GI-DSTS systems have the potential to mitigate approximately 271,097.9 kg of CO_2 throughout their operational lifespan. The net CO_2 mitigation achieved by C-DSTS and GI-DSTS were determined to be 180,704.32 kg and 206,921.09 kg, respectively. Fig. 9(a) and **(b)** show the net CO2

Table 9
Environmental analysis of DSTS.

Parameters	C-DSTS	GI-DSTS
CO_2 emitted (kg/year)	85,935	59,722.25
CO2 mitigation (kg/year)	271,097.9	271,097.9
Net CO2 mitigation (lifetime) (kg)	180,704.32	206,921.09
Carbon credit earned (USD)	4165.37	4194.12

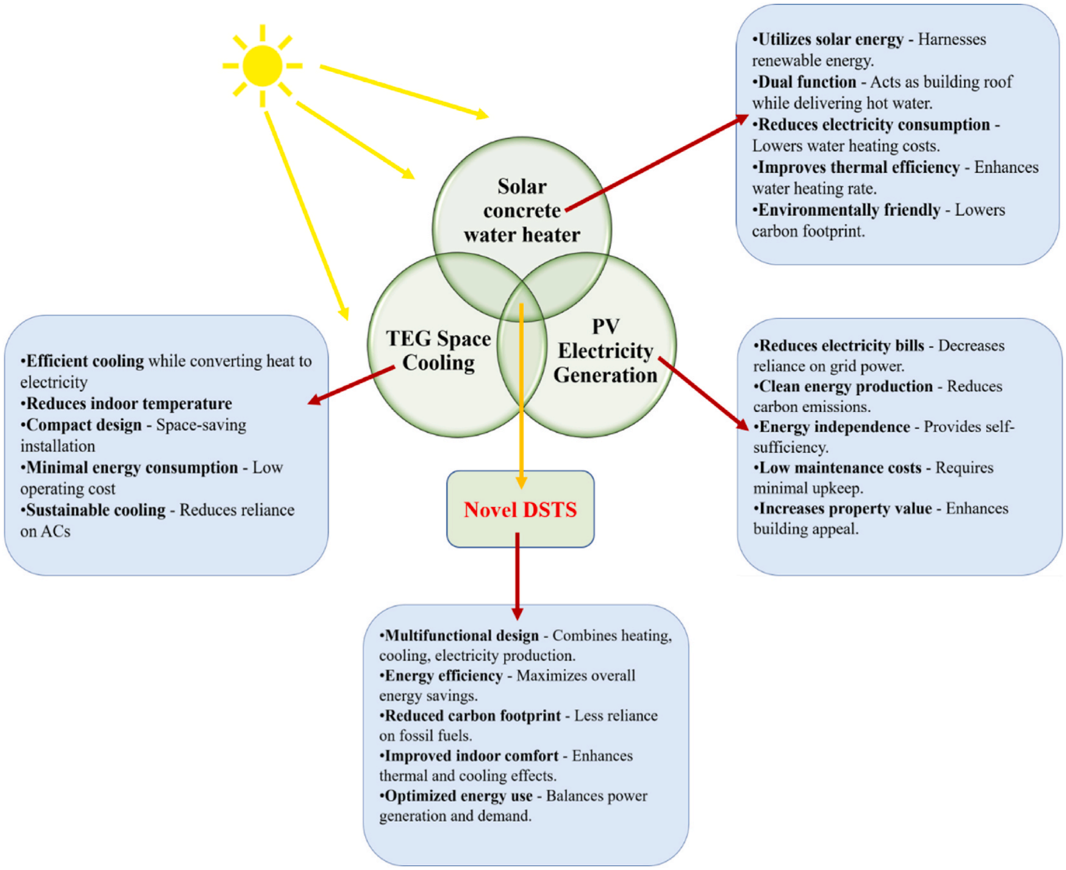

Fig. 11. DSTS advantages.

mitigation of C-DSTS and GI-DSTS for 25 years. This mitigation capacity depends significantly on the operational time of the system, as consistent energy generation maximizes CO_2 reductions. At an estimated carbon reduction price of 23 USD per ton, the carbon credits earned by the C-DSTS and GI-DSTS are 4165.37 USD and 4194.12 USD, respectively. Fig. 10(a) and **(b)** represent the carbon credits earned by both DSTS for 25 years. The results of the environmental analysis are summarized in Table 9. Compared to existing studies, the proposed DSTSs demonstrated superior environmental performance [73,77,78]. Both systems showed a minor decline in annual energy production and CO_2 mitigation over time, likely due to gradual wear and environmental influences. However, they maintain steady financial benefits, reflecting sustainable performance over time. In conclusion, while both C-DSTS and GI-DSTS are highly effective for energy generation and CO_2 mitigation, GI-DSTS outperforms C-DSTS in terms of environmental and financial metrics. Its lower emissions and higher net CO_2 mitigation make it a more sustainable choice over a 25-year operational period.

6. Conclusion

This study comprehensively evaluated copper- and galvanized iron-based Domestic Solar Tri-Generation Systems (C-DSTS and GI-DSTS) that integrate solar water heating, photovoltaic power generation, and thermoelectric cooling within a single concrete slab designed to function as part of a building roof, whose advantages are depicted in Fig. 11. Under the climatic conditions of Vellore, India, the systems demonstrated promising performance, achieving thermal efficiencies of ~24 %, exergy efficiencies up to 37 %, and overall system efficiency of 46.4 %. The C-DSTS and GI-DSTS reduced indoor temperatures by 6.2 °C and 5.9 °C, respectively, while reliably heating 108 L of water to 50 °C within 5 h and generating an annual electricity output of 198 kWh. Techno-economic analysis confirmed their economic viability, with payback periods of 8.2 years (C-DSTS) and 5.3 years (GI-DSTS), and lifetime savings exceeding USD 55,000. Environmentally, the systems mitigated 180,000–207,000 kg of CO_2 emissions, with carbon credit earnings above USD 4000. Beyond performance metrics, the experimental design directly reflects building integration: the reinforced concrete slab replicates a typical roof, enabling dual use of building envelope space, while embedded pipes and TEGs provide hot water and passive indoor cooling without external power input. This multifunctional design enhances occupant comfort, reduces reliance on conventional energy systems, and offers scalability for both new construction and selective retrofitting. While the present study focused on one climatic region, future multi-location validation will be essential to assess adaptability under diverse conditions such as low solar insolation, high humidity, and seasonal extremes. Retrofitting challenges—including structural adjustments, plumbing and electrical integration, and roof-space limitations—also warrant further exploration. Overall, the proposed DSTS represents a practical, durable, and sustainable solution for residential buildings, advancing the transition toward energy-efficient and environmentally responsible construction practices.

While the findings of this study demonstrate the promising performance of the DSTS under the climatic conditions of Vellore, India characterized by high solar irradiance and elevated ambient temperatures the performance metrics may vary across other geographic regions. In particular, lower solar insolation, higher humidity, and colder climates could influence passive cooling performance, thermal gradients, and overall energy yield. The thermal mass and heat-retention characteristics of the concrete slab may also behave differently under seasonal extremes, thereby affecting system efficiency. Although a detailed investigation of these aspects is beyond the scope of the present work, future multi-location testing and modelling will be essential to validate scalability and adaptability under diverse climatic zones. Furthermore, practical challenges in retrofitting existing buildings with DSTS such as structural modifications, integration with plumbing and electrical systems, and roof-space limitations should also be addressed in future studies to ensure effective real-world implementation.

7. Future scope

Future work will focus on improving the cooling performance of thermoelectric generators (TEGs) through optimization experiments. These experiments will assess the effectiveness of graphene sheets, heat sinks, and their combined application in enhancing cooling efficiency. Furthermore, hydrogen production using a proton exchange membrane (PEM) fuel cell powered by the electricity generated by the system is feasible. This study aims to expand the multifunctional capabilities of the Domestic Solar Tri-Generation System by integrating hydrogen production as an additional sustainable energy solution.

CRediT authorship contribution statement

Balamurali Duraivel: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation. **Natarajan Muthuswamy:** Writing – review & editing, Visualization, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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